Case studies

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Government Initiatives in Nepal

Although Nepal stands at a lower position in the global scenario of **Information and Communication Technology (ICT)**, there have been **noticeable developments** in the ICT sector in recent years.

- **Telecommunication facilities** have improved remarkably.
- **Academic institutions and universities** producing ICT professionals have expanded.
- **Electronic and print media** have extended their access to the general public.
- The use of **Internet, emails, and computers** is gaining popularity.
- Several **e-Government applications** are being introduced.

Among Nepal's major e-Government initiatives:

- **Nepal's e-Government Master Plan**, completed in **November 2006**, laid the foundation for digital governance.
- The **Nepal e-Government Master Plan (NeGMP)** was a **strategic national framework** prepared in **November 2006** to guide Nepal's transition toward **digital governance**.
  It was **developed with the technical assistance of the Korean International Cooperation Agency (KOICA)** and aimed to modernize how the Government of Nepal delivers services using **Information and Communication Technology (ICT)**.
- Another important milestone is the preparation of an **ICT Development Project**, which includes detailed investment proposals for prioritized ICT projects.

The **Government of Nepal** is keen and committed to promoting **e-Government** through the implementation of various:

- **G2G (Government-to-Government)**,
- **G2C (Government-to-Citizen)**, and
- **G2B (Government-to-Business)** projects,
  defined under respective priority areas.

Cyber Laws

Cyber Law is the part of the overall legal system that deals with the **Internet, cyberspace, and digital activities**. It covers several subtopics, including:

- Freedom of expression,
- Access to and usage of the Internet,
- Online privacy, and
- Protection against cybercrimes.

The **first cyber law** was the **Computer Fraud and Abuse Act (CFAA)**, enacted in **1986** in the United States.

- This law prohibits **unauthorized access to computers** and outlines the **punishments** for such offenses.

Why Cyber Laws Are Needed

Like any other law, **cyber laws** are created to:

- Protect **individuals and organizations** on the Internet from malicious activities,
- Prevent **cybercrimes** such as hacking, identity theft, data breaches, and online fraud, and
- Maintain **order, trust, and safety** in cyberspace.

If someone violates a cyber law, it allows another person or organization to take **legal action** against the offender.

Implementation of Land Reforms

Land Reform is a government program aimed at redistributing agricultural land among the landless and improving land ownership systems. It involves reforming the **legal and administrative frameworks** — such as laws, regulations, and rules — to ensure fair and efficient land management.

Nepal has been working on land reform for over **50 years**, but several challenges still remain:

- The **real situation of land ownership** is unclear due to poor record-keeping.

- **Land laws are often ignored or poorly enforced.**
- **Multiple ownership disputes** exist, with overlapping land claims.
- **Public trust** in the government's ability to carry out fair land reforms has declined.

Human Resource Management Software (HRMS)

Human Resource Management Software (HRMS) is a digital tool used by organizations to manage employee-related processes throughout their employment lifecycle.
It helps in:

- **Recruitment and candidate management**
- **Employee record management**
- **Payroll and attendance tracking**
- **Performance and workforce optimization**

In **e-Governance**, HRMS supports the government by:

- Increasing work efficiency and productivity
- Saving operational costs
- Reducing human errors
- Ensuring data security
- Supporting better decision-making through digital records

Example: Government offices in Nepal can use HRMS to monitor staff attendance, automate salary distribution, and manage employee data securely.

E-Governance Initiatives Around India

NICNET (National Informatics Centre Network)

NICNET stands for **National Informatics Centre Network**, developed by the **National Informatics Centre (NIC)**, which functions under the **Ministry of Electronics and Information Technology (Government of India)**.
NICNET provides ICT infrastructure and networking to deliver government services online and support **Digital India** initiatives.
Main services include:

- Government Local Area Networks (LAN)
- Video conferencing and secure email
- Remote sensing and GIS support
- Domain registration
- National Cloud and Data Centers
- Cybersecurity services

Example: NICNET helps connect central and state government offices across India for faster communication and coordination.

Collectorate

A **Collectorate** is the **district-level administrative office complex** where all important government offices are located. It is headed by the **District Collector**, who is the chief administrative and revenue officer of the district.
Roles of the District Collector:

- Coordinates all government departments in the district
- Supervises law and order through the police administration
- Oversees **revenue collection** and taxation
- Leads **disaster management and relief operations**
- Can **impose curfew** in emergencies to maintain peace and order

Example: During a flood or earthquake, the District Collector manages relief distribution, rescue operations, and coordination between police, army, and local bodies.

Computer-Aided Administration of Registration Department (CARD)

The **CARD project** was launched in **Andhra Pradesh, India**, to completely **computerize the land registration process**. Before this system, property registration was slow, manual, and prone to errors. The CARD project modernized this system by introducing **transparent, quick, and reliable** digital operations.
Key benefits of CARD:

- **Faster processing:** Registration that once took hours or days can now be completed within minutes.

- **Transparency:** Property valuation and stamp duty calculation are automated and standardized.
- **Simplified process:** Citizens can easily retrieve copies of registered documents instantly.
- **Efficiency:** The system improved accuracy and eliminated manual accounting delays.

Example: A citizen in Andhra Pradesh can now register property, pay duties, and get ownership documents digitally—without needing to go through middlemen or manual paperwork.

Smart Nagarpalika (Smart Municipality)

Smart Nagarpalika is a **national-level e-Governance project** developed by the **National Informatics Centre (NIC)** under the **Ministry of Communication and Information Technology, India**.
It focuses on **computerizing and modernizing urban local bodies** (Municipalities) to make city management efficient, transparent, and citizen-friendly.
Main functions covered under Smart Nagarpalika:

- Birth and death registrations
- Property tax collection
- Water tap connection management and billing
- Financial accounting and auditing
- Building permission approvals
- Monitoring court cases and legal matters

A detailed **System Requirements Specification (SRS)** was prepared to develop software for these municipal functions.
Example: Citizens can apply for building permits, pay water bills, and register births or deaths online without visiting the municipal office.

National Reservoir Level and Capacity Monitoring System

A **reservoir** is an artificial water body created by constructing a **dam**, used for purposes such as **hydroelectric power generation, irrigation, flood control, water supply, and navigation**.
The **National Reservoir Level and Capacity Monitoring System** is a **network-based application** designed to monitor water levels, storage capacities, and usage in reservoirs across states and at the national level.
Purpose and benefits:

- Ensures **efficient water utilization** and crop planning.
- Helps estimate **hydropower generation capacity**.
- Maintains **uniform communication protocols** between central and state authorities.
- Provides **early warning signals** for potential problems like droughts, floods, or low crop yield.

Example: If water levels drop below normal in several reservoirs, it alerts planners early about possible irrigation shortages.

Computerization in Andhra Pradesh

The **Andhra Pradesh State Trading Corporation (APSTC)** underwent a complete digital transformation carried out by the **National Informatics Centre (NIC), Hyderabad**.
This project included:

- Hardware and network setup
- System study and analysis
- Software design and development
- Staff training and support

Outcome:
The APSTC's operations became faster, more reliable, and transparent. The system now manages trading data, accounting, and reporting digitally.
Example: APSTC can now track stock levels, sales, and supplier information in real-time using a computerized system.

Ekal Seva Kendra

Ekal Seva Kendra is an **e-Governance initiative** launched in **Haryana, India**, to provide multiple government services to citizens from **a single window (one-stop center)**.
Services provided include:

- Driving and conductor licenses
- Vehicle registration
- Caste and residence certificates
- Birth and death certificates

Objectives of Ekal Seva Kendra:

- Deliver services **on time**
- Provide **a single point of contact** for citizens
- Simplify government procedures
- Ensure **transparency and accountability**
- Create a **self-sustained, citizen-friendly system**

Example: A person can visit Ekal Seva Kendra to apply for both a driving license and a caste certificate at the same place, without having to visit different offices.

Sachivalaya Vahini (E-Governance in Secretariat)

- **Purpose:** To computerize internal administrative functions in the state secretariat — the top-level decision-making office.
- **Key Systems:**
    - **Patra (LMS):** Manages and tracks letters received in different departments.
    - **Kadatha (FMS):** Tracks and monitors official files for faster decision-making.
    - **Mokaddame (CCMS):** Monitors court cases where the government is involved.
    - **Aaya-Vaya (BMS):** Manages and monitors the state's budget proposals.
- **Impact:** Improved file tracking, faster decision-making, transparency, and accountability in administrative work.

Bhoomi

- **Purpose:** Computerization of **land records** to help rural farmers access ownership certificates easily.
- **Developed by:** NIC (National Informatics Centre), Bangalore.
- **Features:** Digital land records, quick issue of ownership certificates, reduced corruption.
- **Impact:** Empowered farmers, minimized manipulation in land records, and made record retrieval faster and transparent.

Judiciary (e-Courts Initiative)

- **Purpose:** To introduce IT in court processes for efficiency and transparency.
- **Implemented by:** NIC in 1990 (initially in Supreme Court).
- **Features:**
    - Digital case database.
    - Automated cause lists (daily schedules).
    - Easy grouping of similar cases.
    - Quick retrieval of dismissed or reviewed cases.
- **Impact:** Faster case processing, reduced manual workload, and more transparency in judicial administration.

E-Khazana

- **Purpose:** Online financial management system for handling **budget control, audit, and payments** in government offices.
- **Features:**
    - Entry-level data validation.
    - Budget tracking and pre-audit control.
    - Generation of monthly financial reports.
    - Administrator assigns user roles and oversees expenditures.
- **Impact:** Transparent public fund management, reduced errors, and faster financial processing.

Director General of Foreign Trade (DGFT)

- **Purpose:** To computerize the operations of the DGFT under the Ministry of Commerce to facilitate **foreign trade**.
- **Features:**
    - Online processing of Export-Import (EXIM) applications.
    - Reduced paperwork and physical visits.
    - Ensures clearance of electronic applications within 24 hours.
- **Impact:** Speed, transparency, and modernization of India's foreign trade system.

PRAJA

- **Purpose:** To provide **all government and allied services** to rural citizens through district-level portals.
- **Focus:** Bringing government closer to people and empowering rural communities.
- **Services:** Issuing certificates, providing information on government programs, and enabling both **G2C (Government-to-Citizen)** and **C2C (Citizen-to-Citizen)** services.
- **Impact:** Strengthened rural governance and accessibility of government services.

E-Seva

- **Purpose:** To create a **single-window system** offering a wide range of government services.
- **Services:** Online payment of bills, issuing of certificates, licenses, and permits.
- **Features:** Integrated, transparent, efficient, and reliable service model.
- **Impact:** Saved citizens time and effort by avoiding visits to multiple offices; increased trust in government efficiency.

E-Panchayat

- **Purpose:** Developed by **NIC Hyderabad**, it digitizes operations at the **Gram Panchayat (village government)** level.
- **Features:** Web-enabled, citizen-centric software that manages local governance data and services.
- **Impact:** Empowered rural officials and citizens; simplified administration and improved transparency in village governance.

GISTNIC (General Information Services of NIC)

- **Purpose:** To provide **online general information** to the public and act as a government communication network.
- **Objectives:**
  - Deliver data from government databases.
  - Promote informatics and internet use among citizens.
  - Serve as a communication channel between government departments.
- **Impact:** Helped build an **information culture** and supported digital literacy in early stages of e-Governance in India.

E-Governance Initiatives Around the World

United States (President's Management Agenda – PMA, 2001)

- **Goal:** Make all government services accessible **within 3 clicks** on the internet.
- **Principles:**
  - **Citizen-centric** (focus on people, not bureaucracy).
  - **Result-oriented** (focus on outcomes, not procedures).
  - **Market-based** (focus on efficiency and innovation).
- **Impact:** Guided U.S. digital transformation with a citizen-first approach.

China (Zhongguancun Hi-tech E-Park, Beijing)

- **Purpose:** Use ICT to improve government efficiency and reduce corruption.
- **Features:**
  - Online business registration and licensing.
  - Digital tax filing and financial report submission.
  - Over **6000 enterprises** using online services since 2000.
- **Impact:** Increased speed, reduced corruption, and became a model for national e-Governance expansion.

Brazil (Poupatempo / Time Saver Centers)

- **Purpose:** Provide multiple government services **in one location** for citizens.
- **Services:**
  - Vehicle registration
  - Driving license (new/renewal)
  - National ID card

- Labor card
- Unemployment insurance
- **Impact:** Simplified access, improved public trust, and ensured **equal service standards** for all citizens.

Sri Lanka (Kothmale Radio/Internet Project)

- **Purpose:** Bridge the **digital divide** in rural areas through radio-integrated internet access.
- **Features:**
  - Community radio connected with the internet.
  - Locals could ask questions and get online information via radio hosts.
  - Promoted ICT awareness and local problem-solving.
- **Impact:**
  - Increased ICT literacy.
  - Encouraged youth involvement.
  - Empowered rural communities to find new livelihood solutions